



Electrical Machines-I

ECE-2207

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“Read! In the name of thy Lord
and Cherisher, Who created-”.

[Sura Al-'Alaq]



Study Materials

For Basic Knowledge

- ❑ **Electric Machines -Theory, Operation, Application and Control**
Charles I. Hubert
- ❑ **Electric Machinery Fundamentals**
Stephen J. Chapman

For Govt. Job of BD

- ❑ **A text Book of Electrical Technology (Volume II)**
B. L. Theraza, A. K. Theraza
- ❑ **Principles of Electrical Machines**
V. K. Mehta, Rohit Mehta

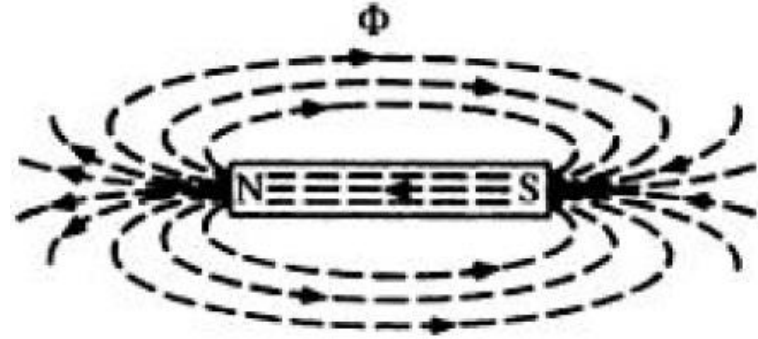
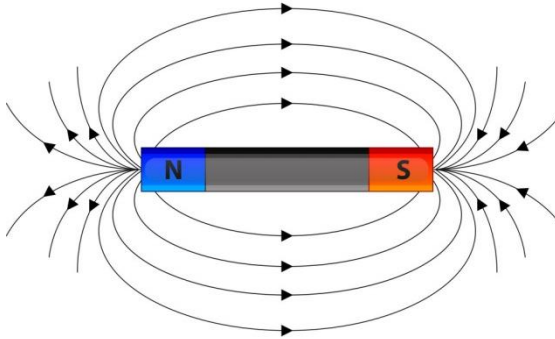
Magnetic Field is the Heart of **Electrical Machines**

**So, Let's see how it can be
generated!**

Magnetic Field

❑ Magnetic field can be generated by

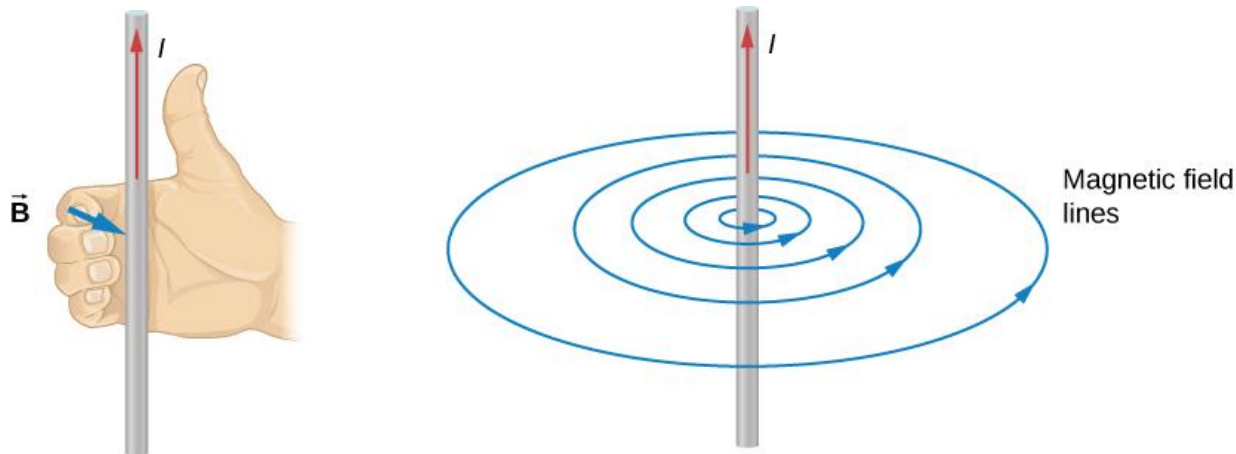
a. **A permanent magnet**



The direction of the magnetic field supplied by a magnet is from the north pole to the south pole but is **south to north within the magnet** itself, i.e., the flux forms a closed loop.

Magnetic Field

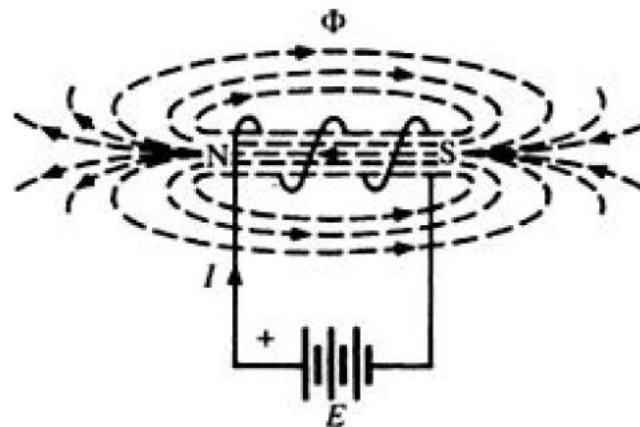
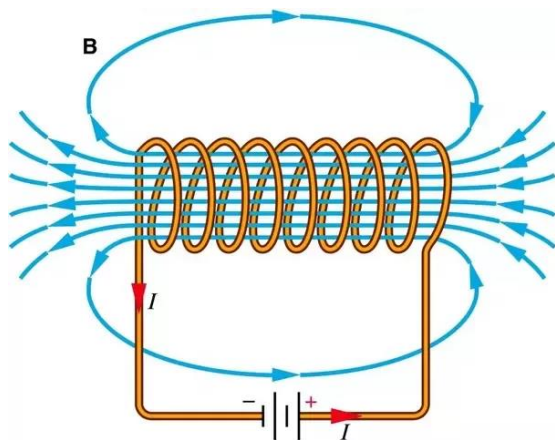
b. **A current** (both static and time varying) **carrying conductor**



The direction of the magnetic field around a current can be determined by the **right-hand rule**: Grasp the conductor with the right hand, with the thumb pointing in the direction of conventional current and the fingers will curl in the direction of the magnetic field.

Magnetic Field

c. **A current** (both static and time varying) **carrying coil of wire**



The direction of the magnetic field generated by a current through a coil of wire is also determined by the **right-hand rule**. Grasp the coil with the right hand, with the finger curled in the direction of the current and the thumb will point in the direction of the magnetic field.



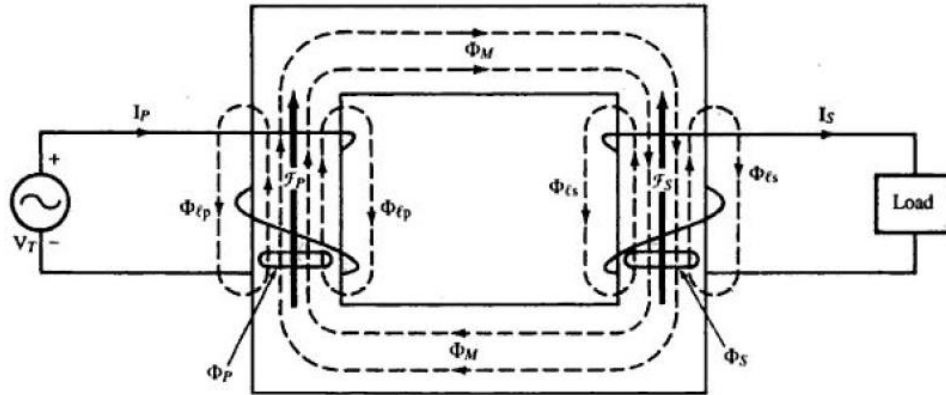
Magnetic Field

❑ **Magnetic fields** are the fundamental mechanism by which **energy is converted** from one form to another in **motors**, **generators** and **transformers**. The basic principles that describe how magnetic fields are used in these devices are:

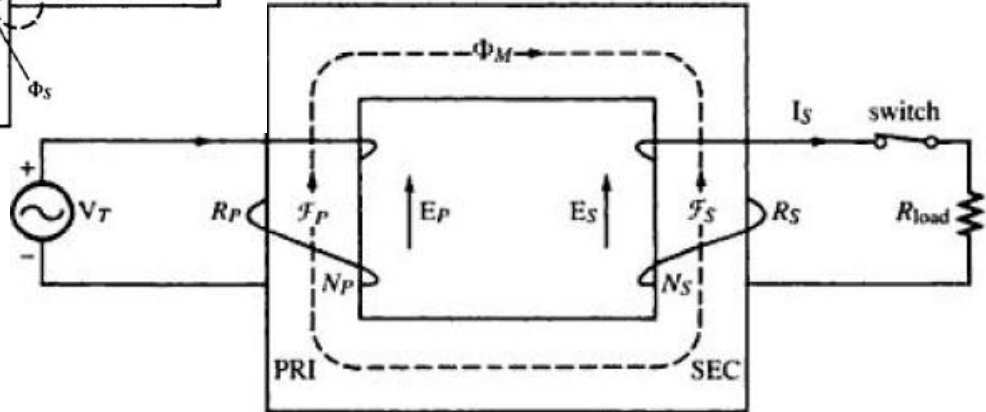
- A time-changing **magnetic field** induces a voltage in a coil of wire if it passes through that coil. **(Basis of transformer action.)**
- A current-carrying wire in the presence of a **magnetic field** has a force induced on it. **(Basis of motor action.)**
- A moving wire in the presence of a **magnetic field** has a voltage induced in it. **(Basis of generator action.)**

Basis of Transformer Action

A time-changing magnetic field induces a voltage in a coil of wire if it passes through that col.

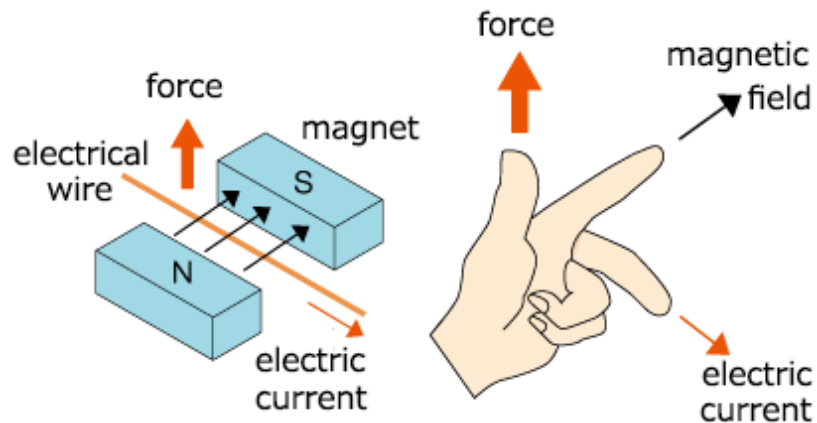


The Direction of the Induced Voltage is determined by **Lenz's Law**



Basis of Motor Action

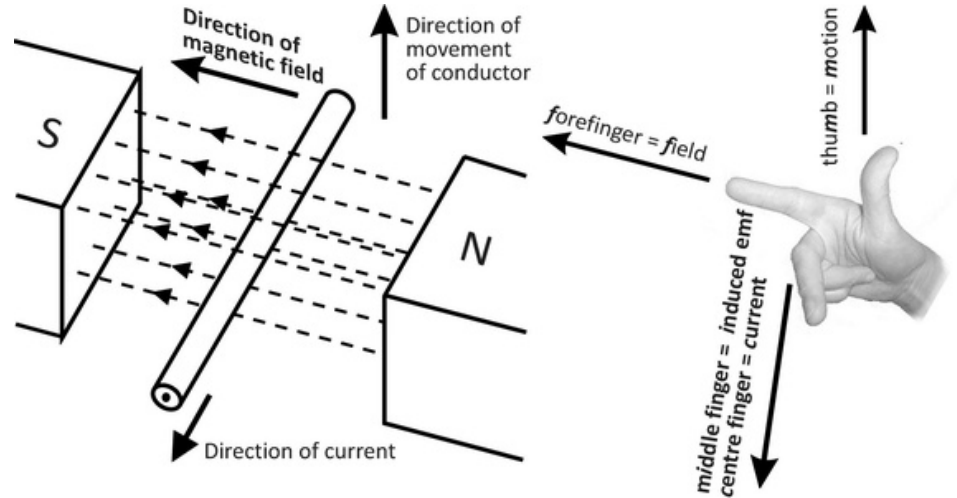
A current-carrying wire in the presence of a magnetic field has a force induced on it.



Fleming's **Left** Hand Rule for Motor

Basis of Generator Action

A moving wire in the presence of a magnetic field has a voltage induced in it.



Fleming's **Right** Hand Rule for Generator